

4. (a) (i) coherent light refers to light wave which possess the same wavelength and frequency, but of a constant phase difference.

(ii) Vapour is made up of atoms or molecules of the same type and the electrons absorb energy and promote to a higher energy level. Since the excited states is unstable, the electrons will 'drop' to a lower energy state after a very short amount of time, releasing the energy in the form of visible light. This process is known as spontaneous emission. As the de-excitation is a random process, the timing of exactly when each electron will de-excite and emit a photon is unpredictable and not synchronised. As such the light emitted by the vapour is not at constant phase difference.

(b) fringe separation = $10.4 \text{ mm}/5 = 2.08 \text{ mm}$
wavelength = 633 nm
distance from slit to screen = 2.95 m

$$x = \frac{\lambda D}{a}$$

$$0.00208 = \frac{(633 \times 10^{-9})(2.95)}{a}$$

$$a = 8.98 \times 10^{-4} \text{ m} = 0.898 \text{ mm}$$